

Liyuan Gao

 gaoliyuan0720@gmail.com  Liyuan Gao  Liyuan Gao
 lily_lygao  <https://liyuan-gao.github.io/>

Research Interests

Explainable AI, Large Language Models, Natural Language Processing, Machine Learning, Applications in Medical Informatics and Bioinformatics.

Education

Aug.2020 – present  **Ph.D., Computer Science, Texa Tech University**
Sep.2016 – Jan.2019  **M.Sc., Computer Technology, University of Science & Technology Beijing**
Aug.2012 – Jun.2016  **B.S., Computer Science and Technology, Yanbian University**

Employment History

Aug.2021 – present  **Teaching and Research Assistant**, Texas Tech University
Jan.2019 – Jul.2021  **Machine Learning Engineer**, Institute of Software Chinese Academy of Sciences
Sep.2016 – Jun.2018  **Teaching Assistant**, University of Science and Technology Beijing

Research Publications

- 1 J. Croft, **L. Gao**, V. Sheng, and J. Zhang, “Deep-learning uncovers certain ccm isoforms as transcription factors,” *Frontiers in Bioscience-Landmark*, vol. 29, no. 2, p. 75, 2024.
- 2 J. Croft, B. Grajeda, **L. Gao**, J. Abou-Fadel, A. Badr, *et al.*, “Whole-genome omics elucidates the role of ccm1 and progesterone in cerebral cavernous malformations within cmpn networks,” *Diagnostics*, vol. 14, no. 17, p. 1895, 2024.
- 3 **L. Gao**, L. Sun, L. Jis, and V. S. Sheng, “Enhancing transcription factor prediction through integrating domain knowledge with logic tensor networks,” in *Proceedings of the AAAI Conference on Artificial Intelligence (submitted to AAAI 2025)*, 2024.
- 4 **L. Gao**, H. Zhan, and V. S. Sheng, “Exploring task-specific dimensions in word embeddings through automatic rule learning,” in *International Conference on Artificial Neural Networks*, Springer, 2024, pp. 199–214.
- 5 **L. Gao**, M. Zhang, and V. S. Sheng, “Enhancing transcription factor prediction through multi-task learning (student abstract),” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 23 500–23 502.
- 6 B. Tang, **L. Gao**, and V. S. Sheng, “Explainable protein-protein interaction prediction using hybrid-feature-enhanced graph attention networks,” in *29th Annual International Conference on Research in Computational Molecular Biology (submitted to RECOMB 2025)*, 2024.
- 7 H. Zhan, **L. Gao**, and V. Sheng, “Privacy-preserving unsupervised spherical text embeddings (accepted),” in *2024 international joint conference on neural networks (IJCNN)*, IEEE, 2024.
- 8 J. Croft, **L. Gao**, O. Quintanar, V. Sheng, and J. Zhang, “Identification of cholangiocarcinoma (cca) subtype-specific biomarkers,” *bioRxiv*, pp. 2023–08, 2023.
- 9 J. Croft, O. Quintanar, **L. Gao**, V. Sheng, and J. Zhang, “Novel hepatocellular carcinomas (hcc) subtype-specific biomarkers,” *bioRxiv*, pp. 2023–08, 2023.

- 10 L. Gao, K. Shu, J. Zhang, and V. S. Sheng, "Explainable transcription factor prediction with protein language models," in *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, IEEE, 2023, pp. 853–856.
- 11 L. Gao, H. Zhan, A. Chen, and V. S. Sheng, "Towards fair and selectively privacy-preserving models using negative multi-task learning (student abstract)," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 16 214–16 215.
- 12 L. Gao, H. Zhan, and V. S. Sheng, "Mitigate gender bias using negative multi-task learning," *Neural Processing Letters*, pp. 1–16, 2023.
- 13 H. Zhan, L. Gao, K. Zhang, Z. Chen, and V. S. Sheng, "Defending the graph reconstruction attacks for simplicial neural networks," in *2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)*, IEEE, 2023, pp. 1–9.
- 14 H. Wang, F. Yuan, L. Gao, R. Huang, and W. Wang, "Wear debris classification and quantity and size calculation using convolutional neural network," in *Cyberspace Data and Intelligence, and Cyber-Living, Syndrome, and Health: International 2019 Cyberspace Congress, CyberDI and CyberLife, Beijing, China, December 16–18, 2019, Proceedings, Part I 3*, Springer, 2019, pp. 470–486.
- 15 H. Wang, R. Huang, L. Gao, W. Wang, A. Xu, *et al.*, "Wear debris classification of steel production equipment using feature fusion and case-based reasoning," *ISIJ International*, vol. 58, no. 7, pp. 1293–1299, 2018.

Projects

- (1) **Clinical Cross-Modality Representation Explanation:** This project merges varied datasets, encompassing biological data and lifestyle information, with associated health outcomes. Our interdisciplinary strategy seeks to develop comprehensive and understandable models capable of predicting health outcomes by integrating biological markers with lifestyle factors. This initiative supports the advancement of personalized medicine and is currently in progress.
- (2) **Explainable Protein-Protein Interaction (PPI) Using Hybrid Feature-Enhanced Dynamic Graph Attention Networks:** We propose an information-dense hybrid feature protein graph representation that incorporates both protein language model features and manually extracted features. This approach employs dynamic graph attention networks (GATv2) for PPI prediction. This method utilizes dynamic graph attention networks (GATv2) to predict protein-protein interactions. Moreover, we have developed advanced explainable AI tools that facilitate the examination and visualization of how various features—ranging from sequence-based to graph-based and structure-based—enhance model interpretability.
- (3) **Advancing Transcription Factor Prediction: Integrating Domain Knowledge with Logic Tensor Networks:** In this project, we introduce an innovative strategy that leverages Logic Tensor Networks (LTNs), a neurosymbolic framework that integrates data and logic in a differentiable first-order logic language. Our methodology uniquely incorporates critical domain-specific knowledge into the TF prediction model, thereby significantly enhancing both the performance and interpretability of the predictions.
- (4) **ESM-TFpredict Model:** We developed the ESM-TFpredict model, combining a pre-trained protein language model with 1-D CNNs for transcription factor prediction. The model employs integrated gradients to elucidate decision-making, highlighting key features in TF identification.
- (5) **Transcription Factor Prediction through Multi-Task Learning:** We introduced a robust multi-task learning framework specifically tailored to harness both TF-specific annotations and TF-related domain annotations, thereby enhancing the accuracy of TF predictions.

- (6) **Word Embedding Explanation:** We introduced a novel methodology to discover task-related dimensions within word embeddings. This project showcased the effectiveness of automatic rule learning in extracting crucial dimensions tailored to specific tasks.
- (7) **Negative Multi-Task Learning:** We introduced a selective privacy-preserving method combined with a negative multi-task learning framework. This innovative approach effectively mitigated gender bias while protecting sensitive information in word embeddings.
- (8) **Privacy-Preserving Spherical Text Embeddings:** We proposed a novel approach: the differential private spherical text generative model (DP-STGM), which facilitates learning text embeddings within the spherical space while ensuring privacy via efficient Riemannian optimization and the framework of differential privacy.
- (9) **Wavefront Restoration:** Focusing on self-adaptive optics, our team utilized the Generative Adversarial Networks to automate wavefront image generation. This significantly enhanced wavefront correction and imaging quality while reducing system complexity.
- (10) **Wear Debris Identification:** We developed a system using CNNs and Faster R-CNN models to classify and quantify wear debris in steel production equipment, incorporating the GrabCut algorithm for improved accuracy.

Skills

Languages	Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	Python, C/C++, Matlab, R.
DL tools	Tensorflow, Pytorch.
Misc.	Academic research, teaching, training, etc.

Miscellaneous Experience

Awards and Achievements

2021	ICDM 2021 student attendance award.
2020-2024	Distinguished Graduate Student Assistantship, Texas Tech University.

Certification

2020	CN Patent. A wavefront restoration system based on generative adversarial network.
2018	Software Copyright. Wear Debris Automatic Recognition System.
	CN Patent. An automatic measurement device for crack tip opening displacement image.
2017	CN Patent. An automatic scoring method and system for operation program.